Project Development
- SpheroidMetrics

Progress update - 10 Nov

Research and Planning

Meet with Dalia to clarify:

- The problem they want solved (make sure that both of our problem statement and objectives align)
- Tangible outputs (e.g.,pipeline, model, report)
- Success criteria (achieving accuracy over 90% for the segmentation and successful metrics analysis.)

Prepare Project Brief

- Background
- Problem statement
- Objectives
- Deliverables
- Timeline overview

Define Roles and Responsibilities

Create a High-Level Plan - and Git tasks management

Learning tools - Git, Roboflow, VScode, Keynotes, Lucidchart

Data collection and Annotation

- Requested fully annotated YOLO-format data from Dalia.
- Converted YOLO annotations to **U-Net compatible masks** and validated to ensure:
  - All images contain annotations.
  - Annotations correctly represent **spheroids only**.
- Created a **Google Drive directory** for organized data storage and easy team access.

- Repeat the same workflow for the next batch of annotated data from Dalia.
- Perform **data augmentation** (horizontal & vertical flips, 180° rotation) to expand the training dataset.

Implementation & Training

Data preprocessing

- Locate images in images folder and masks in labels folder
- Match each image with its mask by filename
- Split dataset into training and validation based on _t suffix
- Resize images and masks to fixed size (squish if needed)
- Normalize image pixels using ImageNet mean and std
- including Gaussian blur (kernel size 25, σ ~ U[0.001, 2.0]), random affine transforms (translations ≤ 25%, rotations ≤ 180◦, scaling 75–125%, shear ≤ 17.5
- Load images and masks in batches for training.

Implementation & Training

Pipeline implementation

- **YOLO**
  - setup the environment,
  - create pipeline
  - Training in progress
- **U-net**
  - setup the environment,
  - Debug and Refactoring the pipeline
  - Train the model on existing data.

Poster

- Finding the suitable tools -
    - Key notes for poster sections Assemble, Lucidchart for creating the flowchart.
- Define the Sections for Poster
    - Problem statement, Objective, Methodology flow chart, project roadmap, (expected) results.
- Prepare the Section contents
- Create methodology flowchart
- Draft the design and template
- Final Alignment and finishing